

Discrete Mathematics Sheet I

Gallo Tennis / to A Mathematical Room

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Problems in Discrete Math from the books: “A Path to Combinatorics For Undergraduates” by Titu Andreescu, “102 Combinatorial Problems” by Titu Andreescu, and “Graph Theory With Applications” by J. A. Bondy. These weeks problems will be about an introduction to counting techniques, inclusion-exclusion principle and graph theory,

Counting Techniques

1. If 8 identical blackboards are to be divided among 4 schools, how many divisions are possible? How many if each school must receive at least 1 blackboard?
2. An elevator starts at the basement with 8 people (not including the elevator operator) and discharges them all by the time it reaches the top floor, number 6. In how many ways could the operator have perceived the people leaving the elevator if all people look alike to him? What if the 8 people consisted of 5 men and 3 women and the operator could tell a man from a woman?
3. We have \$20,000 that must be invested among 4 possible opportunities. Each investment must be integral in units of \$1000, and there are minimal investments that need to be made if one is to invest in these opportunities. The minimal investments are \$2000, \$2000, \$3000, and \$4000. How many different investment strategies are available if
 - (a) an investment must be made in each opportunity?
 - (b) investments must be made in at least 3 of the 4 opportunities?
4. Suppose that 10 fish are caught at a lake that contains 5 distinct types of fish.
 - (a) How many different outcomes are possible, where an outcome specifies the numbers of caught fish of each of the 5 types?

(b) How many outcomes are possible when 3 of the 10 fish caught are trout?

(c) How many when at least 2 of the 10 are trout?

5. Determine the number of functions

$$f : \{1, 2, \dots, 1999\} \rightarrow \{2000, 2001, 2002, 2003\}$$

satisfying the condition that

$$f(1) + f(2) + \dots + f(1999) \text{ is odd.}$$

6. Find the number of two-digit positive integers that are divisible by both of their digits.

7. For any set S , let $|S|$ denote the number of elements in S , and let $n(S)$ be the number of subsets of S , including the empty set and S itself. If A, B , and C are sets for which

$$n(A) + n(B) + n(C) = n(A \cup B \cup C)$$

and

$$|A| = |B| = 100,$$

then what is the minimum possible value of $|A \cap B \cap C|$?

8. For how many pairs of consecutive integers in $\{1000, 1001, 1002, \dots, 2000\}$ is no carrying required when the two integers are added?

9. Each of two boxes contains both black and white marbles, and the total number of marbles in the two boxes is 25. One marble is taken out of each box randomly. The probability that both marbles are black is $\frac{27}{50}$. What is the probability that both marbles are white?

10. There are ten girls and four boys in Mr. Fat's combinatorics class. In how many ways can these students sit around a circular table such that no boys are next to each other?

11. Find the number of ordered triples of sets (A, B, C) such that

$$A \cup B \cup C = \{1, 2, \dots, 2003\} \quad \text{and} \quad A \cap B \cap C = \emptyset.$$

12. Compute the number of sets of three distinct elements that can be chosen from the set $\{2^1, 2^2, 2^3, \dots, 2^{2000}\}$ such that the three elements form an increasing geometric progression.

13. Let n be an integer greater than four, and let $P_1P_2\ldots P_n$ be a convex n -sided polygon. Zachary wants to draw $n - 3$ diagonals that partition the region enclosed by the polygon into $n - 2$ triangular regions and that may intersect only at the vertices of the polygon. In addition, he wants each triangular region to have at least one side that is also a side of the polygon. In how many different ways can Zachary do this?
14. How many five-digit numbers are divisible by three and also contain 6 as one of their digits?
15. Two squares on an 8×8 chessboard are called *touching* if they have at least one common vertex. Determine if it is possible for a king to begin in some square and visit all the squares exactly once in such a way that all moves except the first are made into squares touching an even number of squares already visited.
16. A total of 119 residents live in a building with 120 apartments. We call an apartment *overpopulated* if there are at least 15 people living there. Every day the inhabitants of an overpopulated apartment have a quarrel and each goes off to a different apartment in the building (so they can avoid each other). Is it true that this process will necessarily be completed someday?

Graph theory

1. Show that two simple graphs G and H are isomorphic if and only if there is a bijection $\theta : V(G) \rightarrow V(H)$ such that $uv \in E(G)$ if and only if $\theta(u)\theta(v) \in E(H)$.
2. Let G be simple. Show that $\varepsilon = \binom{\nu}{2}$ if and only if G is complete.
3. Show that
 - (a) $\varepsilon(K_{m,n}) = mn$;
 - (b) if G is simple and bipartite, then $\varepsilon \leq \nu^2/4$.
4. A k -partite graph is one whose vertex set can be partitioned into k subsets so that no edge has both ends in any one subset; a complete k -partite graph is one that is simple and in which each vertex is joined to every vertex that is not in the same subset. The complete m -partite graph on n vertices in which each part has either $\lfloor n/m \rfloor$ or $\lceil n/m \rceil$ vertices is denoted by $T_{m,n}$. Show that
 - (a) $\varepsilon(T_{m,n}) = \binom{n-k}{2} + (m-1)\binom{k+1}{2}$, where $k = \lfloor n/m \rfloor$;
 - (b) if G is a complete m -partite graph on n vertices, then $\varepsilon(G) \leq \varepsilon(T_{m,n})$, with equality only if $G \cong T_{m,n}$.

5. The k -cube is the graph whose vertices are the ordered k -tuples of 0's and 1's, two vertices being joined if and only if they differ in exactly one coordinate. (The graph shown in figure 1.4b is just the 3-cube.) Show that the k -cube has 2^k vertices, $k2^{k-1}$ edges and is bipartite.
6. (a) The complement G^e of a simple graph G is the simple graph with vertex set V , two vertices being adjacent in G^e if and only if they are not adjacent in G . Describe the graphs K_n^e and $K_{m,n}^e$.
 (b) A simple graph G is self-complementary if $G \cong G^e$. Show that if G is self-complementary, then $\nu \equiv 0, 1 \pmod{4}$.
7. The k -cube is the graph whose vertices are the ordered k -tuples of 0's and 1's, two vertices being joined if and only if they differ in exactly one coordinate. (The graph shown in figure 1.4b is just the 3-cube.) Show that the k -cube has 2^k vertices, $k2^{k-1}$ edges and is bipartite.
8. (a) The complement G^* of a simple graph G is the simple graph with vertex set V , two vertices being adjacent in G^* if and only if they are not adjacent in G . Describe the graphs K_n^* and $K_{m,n}^*$.
 (b) A simple graph G is self-complementary if $G \cong G^*$. Show that if G is self-complementary, then $\nu \equiv 0, 1 \pmod{4}$.
9. An *automorphism* of a graph is an isomorphism of the graph onto itself.
 - (a) Show, using exercise 1.2.5, that an automorphism of a simple graph G can be regarded as a permutation on V which preserves adjacency, and that the set of such permutations forms a group $\Gamma(G)$ (the *automorphism group* of G) under the usual operation of composition.
 - (b) Find $\Gamma(K_n)$ and $\Gamma(K_{m,n})$.
 - (c) Find a nontrivial simple graph whose automorphism group is the identity.
 - (d) Show that for any simple graph G , $\Gamma(G) = \Gamma(G^*)$.
 - (e) Consider the permutation group Λ with elements $(1)(2)(3)$, $(1, 2, 3)$ and $(1, 3, 2)$. Show that there is no simple graph G with vertex set $\{1, 2, 3\}$ such that $\Gamma(G) = \Lambda$.
 - (f) Find a simple graph G such that $\Gamma(G) = \Lambda$. (Frucht, 1939 has shown that every abstract group is isomorphic to the automorphism group of some graph.)
10. A simple graph G is *vertex-transitive* if, for any two vertices u and v , there is an element g in $\Gamma(G)$ such that $g(u) = v$; G is *edge-transitive* if, for any two edges $\{u_1, v_1\}$ and $\{u_2, v_2\}$, there is an element h in $\Gamma(G)$ such that $h(\{u_1, v_1\}) = \{u_2, v_2\}$. Find
 - (a) a graph which is vertex-transitive but not edge-transitive;
 - (b) a graph which is edge-transitive but not vertex-transitive.

11. What is the number of edges in a K^n ?
12. Let $d \in \mathbb{N}$ and $V := \{0, 1\}^d$; thus, V is the set of all 0-1 sequences of length d . The graph on V in which two such sequences form an edge if and only if they differ in exactly one position is called the *d-dimensional cube*. Determine the average degree, number of edges, diameter, girth and circumference of this graph. (Hint for the circumference: induction on d .)
13. Let G be a graph containing a cycle C , and assume that G contains a path of length at least k between two vertices of C . Show that G contains a cycle of length at least $\sqrt{k}$.
14. Is the bound in Proposition 1.3.2 best possible?
15. Let v_0 be a vertex in a graph G , and $D_0 := \{v_0\}$. For $n = 1, 2, \dots$ inductively define $D_n := N_G(D_0 \cup \dots \cup D_{n-1})$. Show that $D_n = \{v \mid d(v_0, v) = n\}$ and $D_{n+1} \subseteq N(D_n) \subseteq D_{n-1} \cup D_{n+1}$ for all $n \in \mathbb{N}$.
16. Show that $\text{rad}(G) \leq \text{diam}(G) \leq 2 \text{rad}(G)$ for every graph G .
17. Prove the weakening of Theorem 1.3.4 obtained by replacing average with minimum degree. Deduce that $|G| \geq n_0(d/2, g)$ for every graph G as given in the theorem.
18. Show that graphs of girth at least 5 and order n have a minimum degree of $o(n)$. In other words, show that there is a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(n)/n \rightarrow 0$ as $n \rightarrow \infty$ and $\delta(G) \leq f(n)$ for all such graphs G .
19. Show that every connected graph G contains a path or cycle of length at least $\min\{2\delta(G), |G|\}$.
20. Show that a connected graph of diameter k and minimum degree d has at least about $kd/3$ vertices but need not have substantially more.